

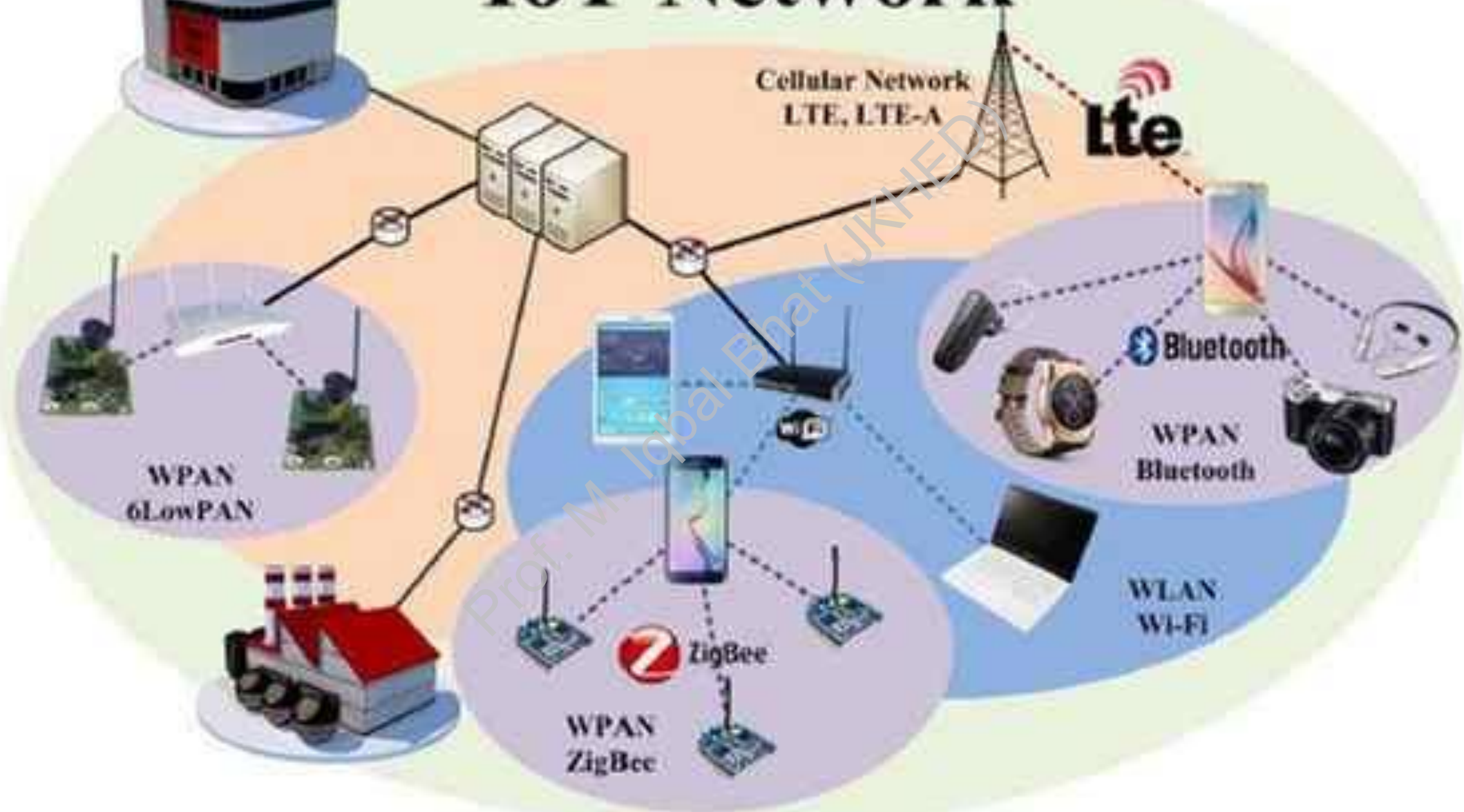
Wireless IoT Network Protocols



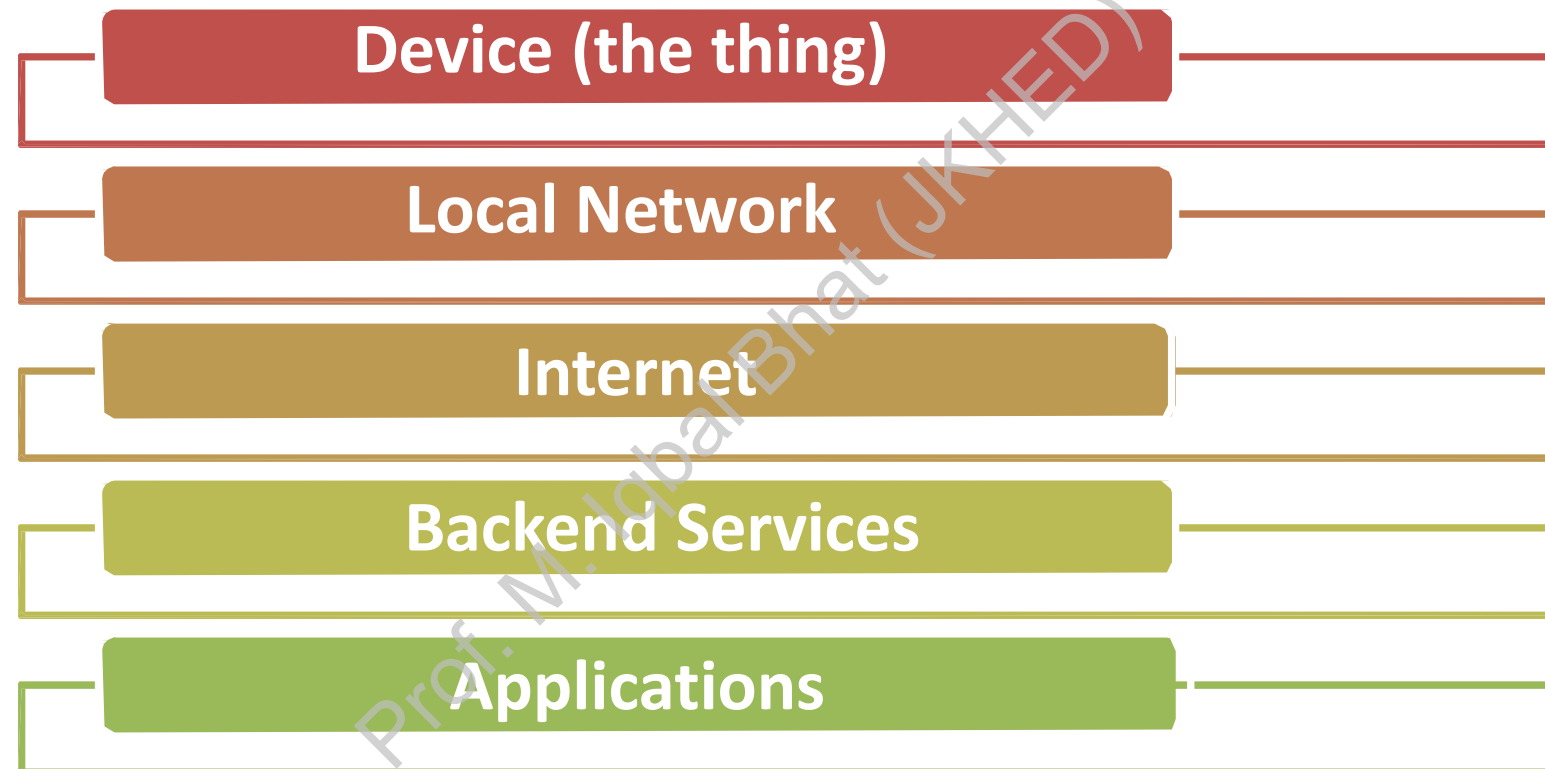
IoT Networking

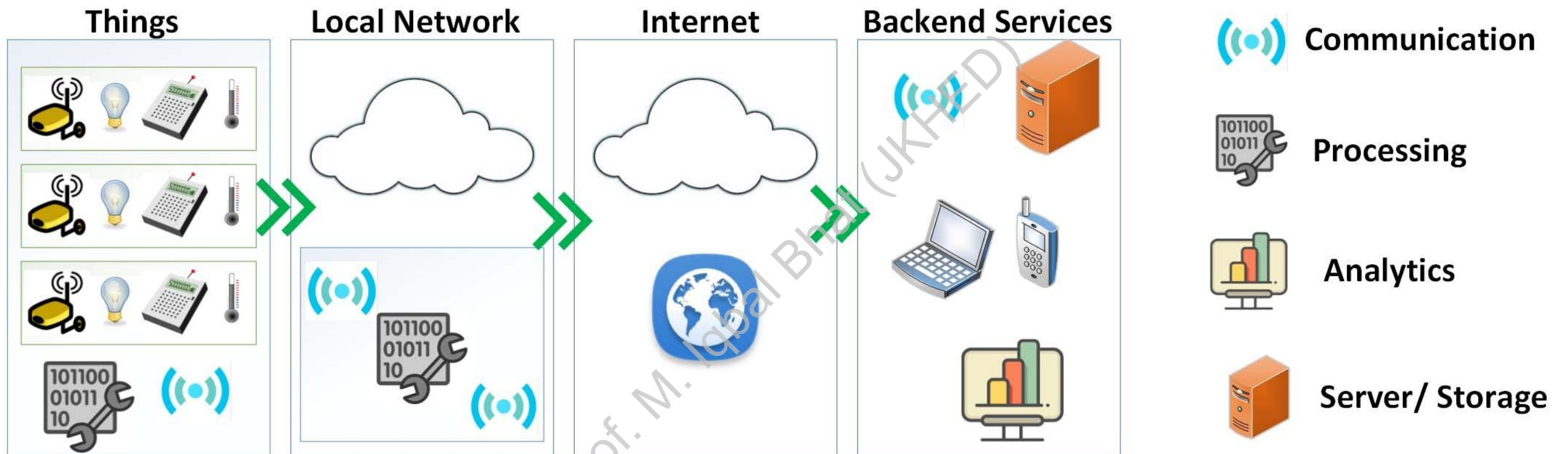
Prof. Mohd. Iqbal Bhat
Government Degree College Beerwah

IoT Network



IoT Components



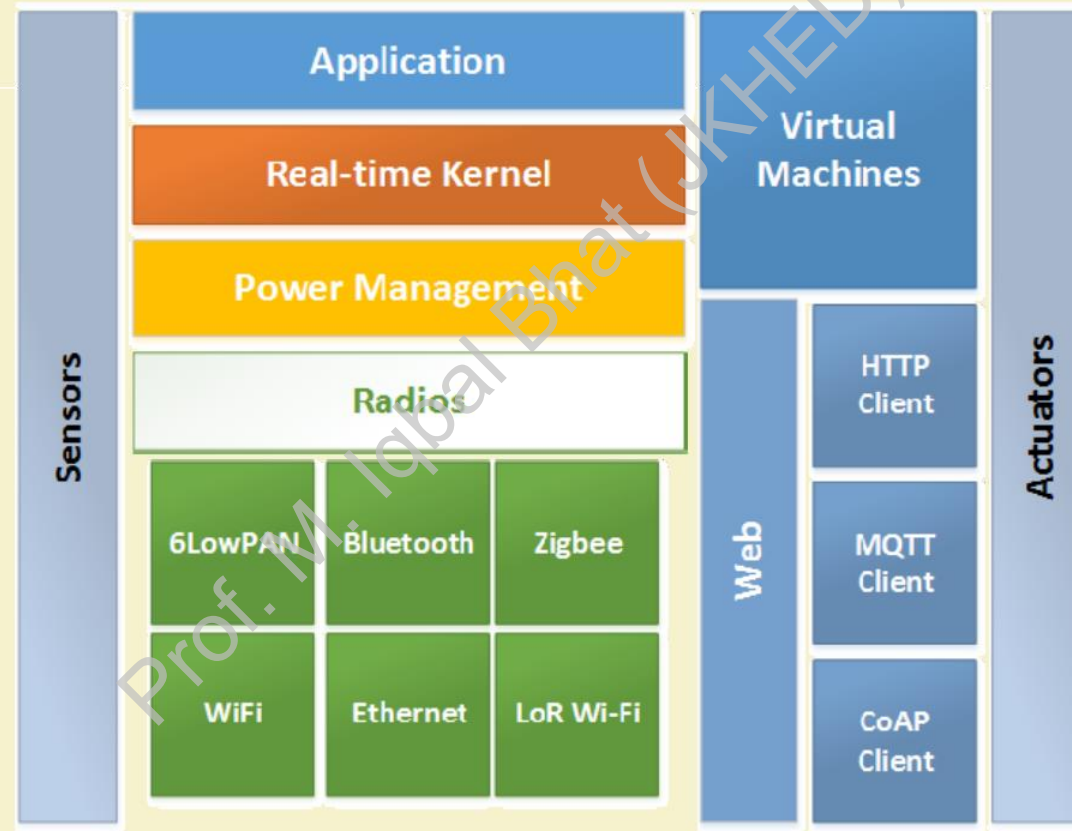


Functional Components of IoT

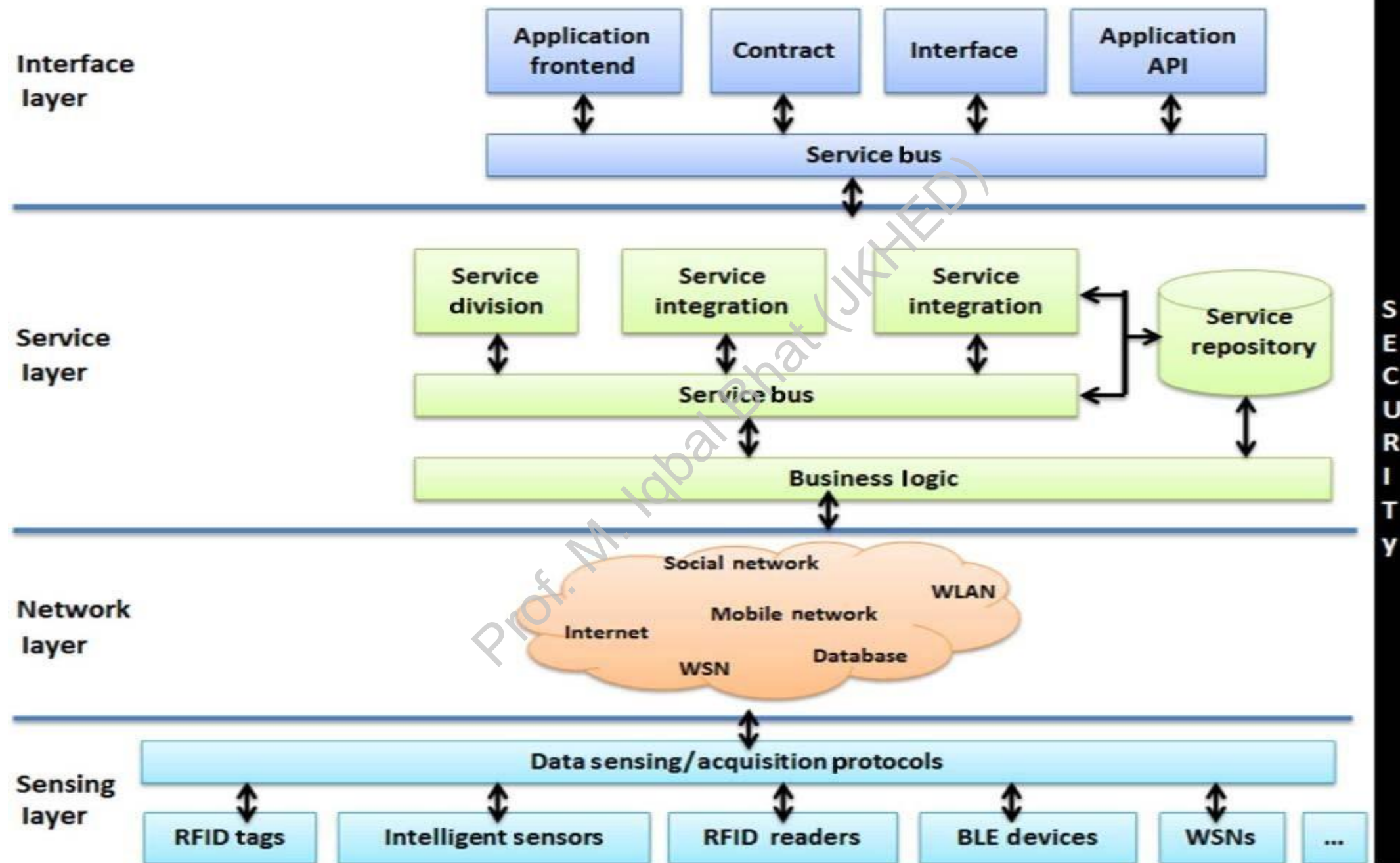
- ✓ Component for interaction and communication with other IoT devices
- ✓ Component for processing and analysis of operations
- ✓ Component for Internet interaction
- ✓ Components for handling Web services of applications
- ✓ Component to integrate application services
- ✓ User interface to access IoT

Source: O Vermesan, P. Friess, “Internet of Things – Converging Technologies for Smart Environments and Integrated Ecosystems”, River Publishers, Series in Communications, 2013

IoT Interdependencies



IoT Service Oriented Architecture



IoT Categories

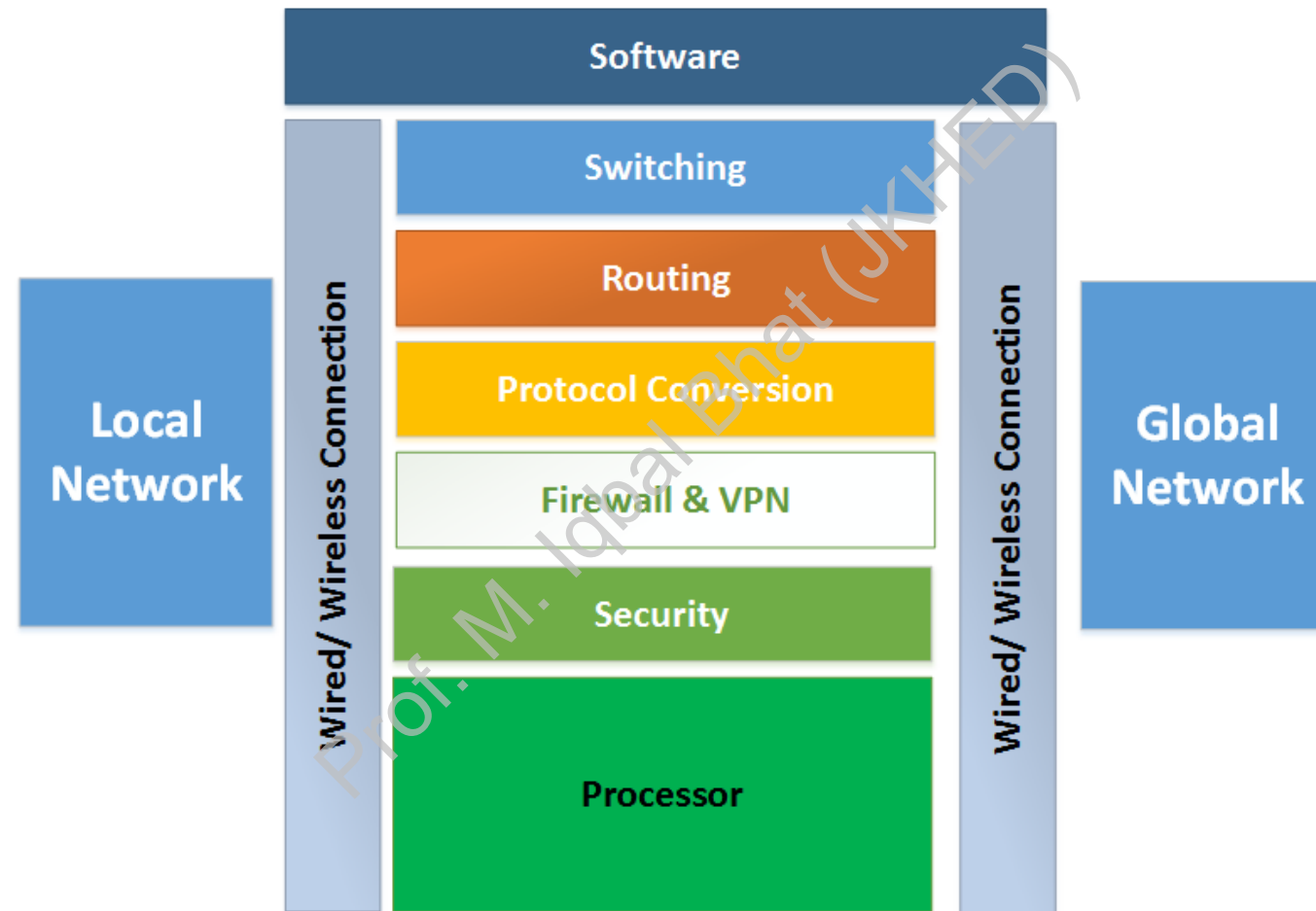
✓ Industrial IoT

- IoT device connects to an IP network and the global Internet.
- Communication between the nodes done using regular as well as industry specific technologies.

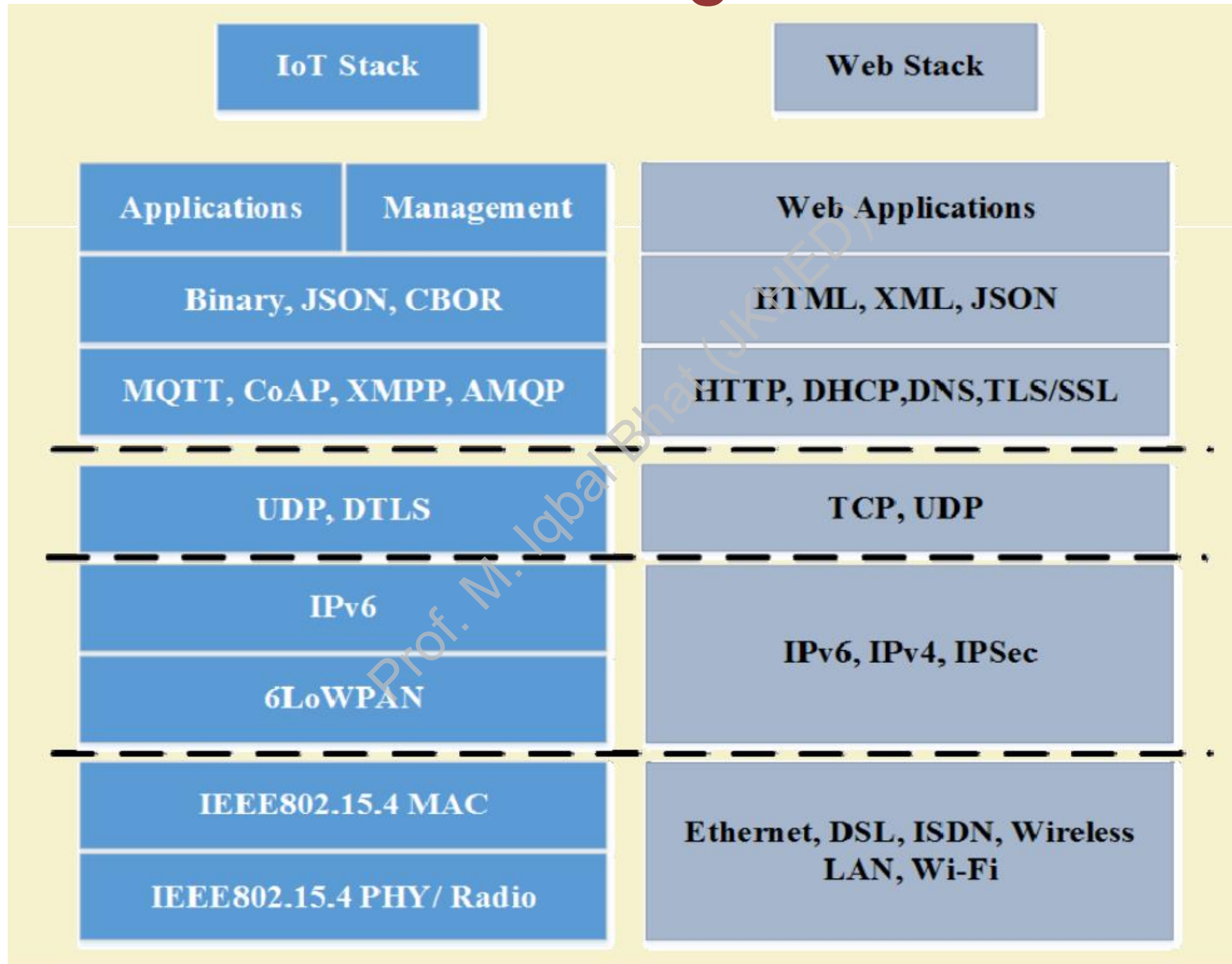
✓ Consumer IoT

- IoT device communicates within the locally networked devices.
- Local communication is done mainly via Bluetooth, Zigbee or WiFi.
- Generally limited to local communication by a Gateway

IoT Gateways



Technical Deviations from Regular Web



Considerations

- ✓ Communication between the IoT device(s) and the outside world dictates the network architecture.
- ✓ Choice of communication technology dictates the IoT device hardware requirements and costs.
- ✓ Due to the presence of numerous applications of IoT enabled devices, a single networking paradigm not sufficient to address all the needs of the consumer or the IoT device.

Complexity of Networks

- ✓ Growth of networks
- ✓ Interference among devices
- ✓ Network management
- ✓ Heterogeneity in networks
- ✓ Protocol standardization within networks

Source: O Vermesan, P. Friess, “Internet of Things – Converging Technologies for Smart Environments and Integrated Ecosystems”, River Publishers, Series in Communications, 2013

Communication Protocols

The following communication protocols have immediate importance to consumer and industrial IoTs:

- ✓ IEEE 802.15.4
- ✓ Zigbee
- ✓ 6LoWPAN
- ✓ Wireless HART
- ✓ Z-Wave
- ✓ ISA 100
- ✓ Bluetooth
- ✓ NFC
- ✓ RFID

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ZigBee[®]

Prof. M. Iqbal Bhat (UKHED)

Zigbee



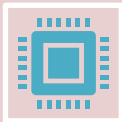
ZigBee is a wireless technology used for small scale projects.



ZigBee is based on the IEEE 802.15.2 protocol.



We use this technology over a small range, that is, 10-100m.



This technology is operating at 2.4GHz and exchanges data at low data rates, that is, 250kbps over a short area.



This technology has various applications in agriculture, automotive sensing, smart homes, remote control, and many more

Features of ZigBee

- ✓ Most widely deployed enhancement of IEEE 802.15.4.
- ✓ The ZigBee protocol is defined by **layer 3 and above**. It works with the 802.15.4 layers 1 and 2.
- ✓ The standard uses layers 3 and 4 to define additional communication enhancements.
- ✓ These enhancements include authentication with valid nodes, encryption for security, and a data routing and forwarding capability that enables mesh networking.
- ✓ The most popular use of ZigBee is wireless sensor networks using the mesh topology.

Source: L.Fenzel, [“What’s The Difference Between IEEE 802.15.4 And ZigBee Wireless?”](#), Electronic Design (Online), Mar. 2013

ZigBee Node Types

✓ *ZigBee Coordinator (ZC):*

- The Coordinator forms the root of the ZigBee network tree and might act as a bridge between networks.
- There is a single ZigBee Coordinator in each network, which originally initiates the network.
- It stores information about the network under it and outside it.
- It acts as a Trust Center & repository for security keys.

Sources:

- "[Wireless Sensor Networks Research Group](#)". Sensor-networks.org. 2010-04-15.
- "[Wireless Sensor Networks Research Group](#)". Sensor-networks.org. 2009-02-05.

ZigBee Node Types

✓ *ZigBee Router (ZR):*

- Capable of running applications, as well as relaying information between nodes connected to it.

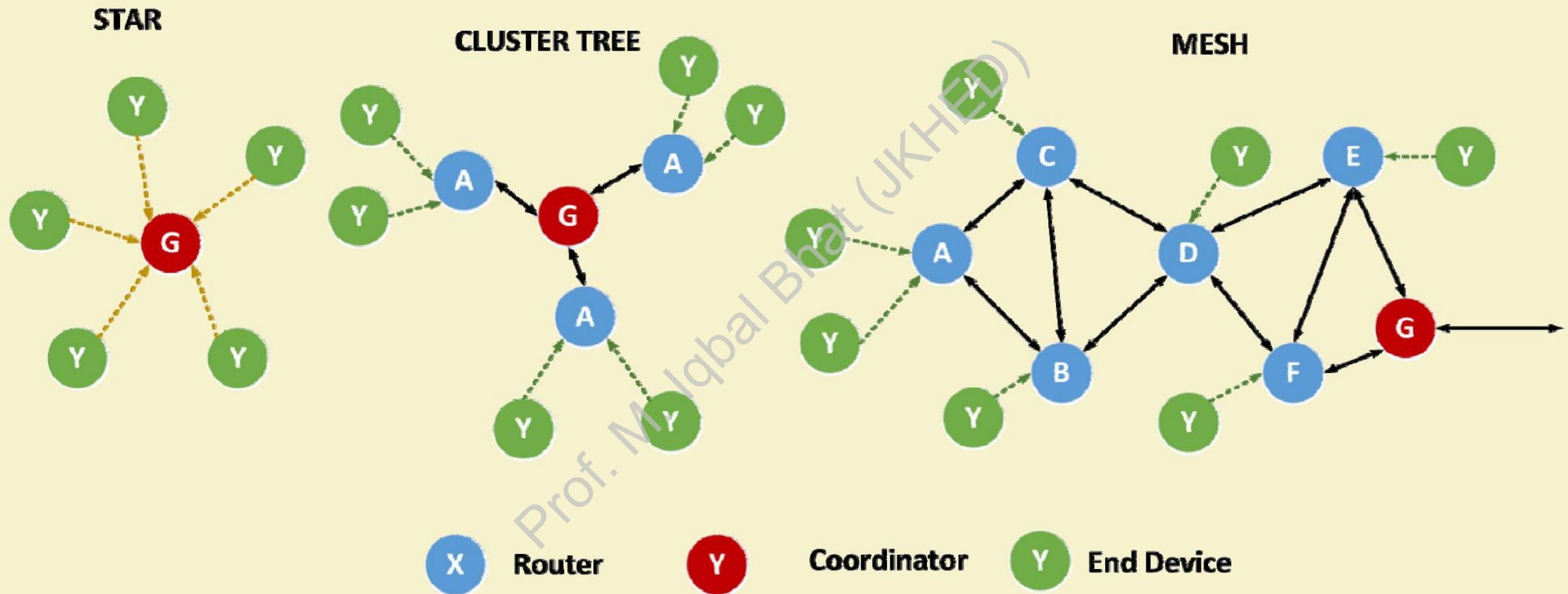
✓ *ZigBee End Device (ZED):*

- It contains just enough functionality to talk to the parent node, and it cannot relay data from other devices.
- This allows the node to be asleep a significant amount of the time thereby enhancing battery life.
- Memory requirements and cost of ZEDs are quite low, as compared to ZR or ZC.

Sources:

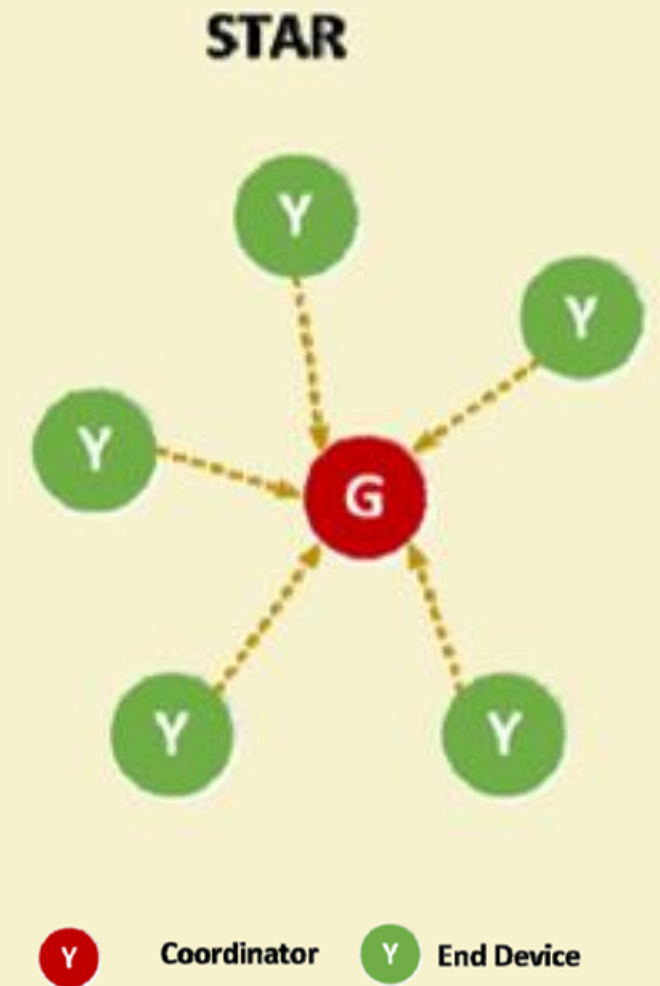
- "[Wireless Sensor Networks Research Group](http://www.sensor-networks.org)". Sensor-networks.org. 2010-04-15.
- "[Wireless Sensor Networks Research Group](http://www.sensor-networks.org)". Sensor-networks.org. 2009-02-05.

ZigBee Topologies



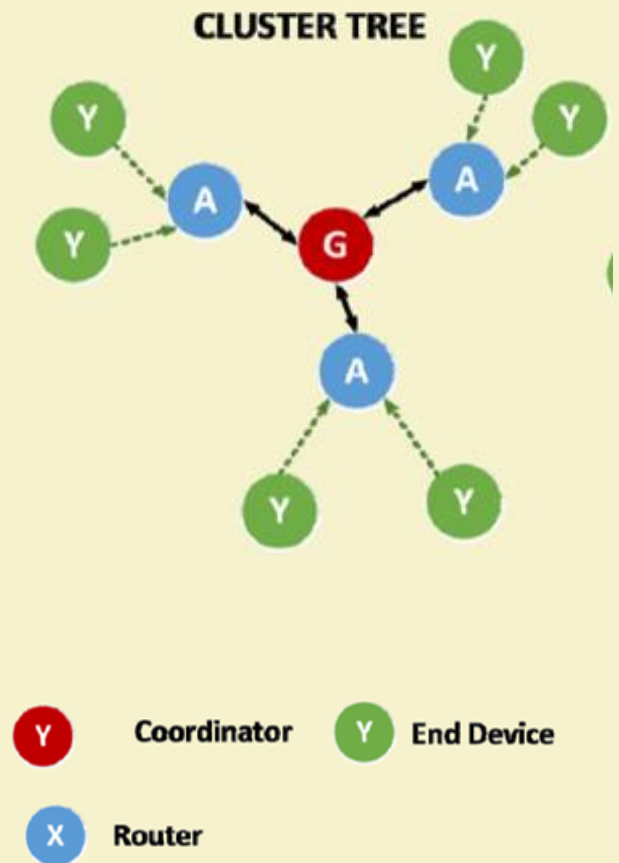
ZigBee Star

- In star network we are having ZigBee coordinator at the center from where the communication starts and all the devices are connected to it.
- Each ZigBee device communicates with the coordinator. If any device wants to send any packet to another device, then it must go through the coordinator, two devices can connect only through the coordinator.
- We use this topology in personal computers, toys, home automation etc.



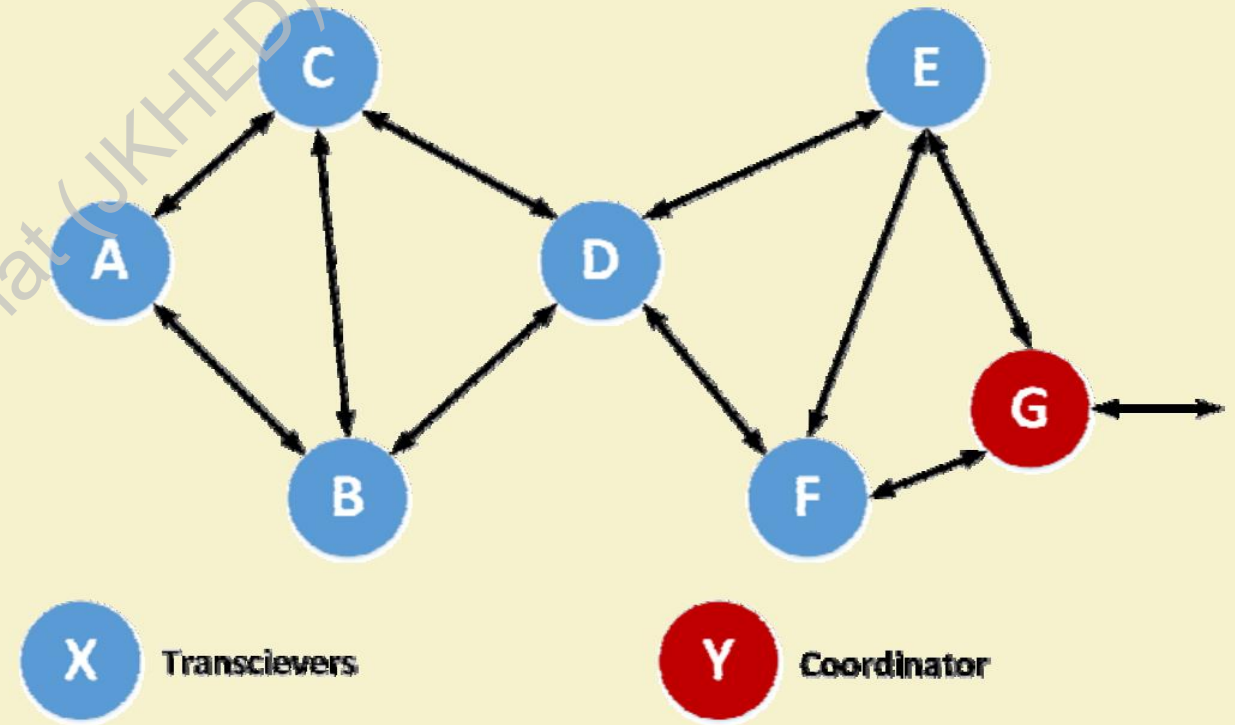
ZigBee Cluster

- In a cluster tree network, we have central coordinate, and nodes that are connected to this center node are known as children.
- Parent with children makes cluster, and there is number of clusters in this topology.
- Communication is only possible if two nodes are directly connected; no two nearby nodes communicate directly.
- Each cluster must have their ID, that is, Cluster ID.
- This topology covers a large area.



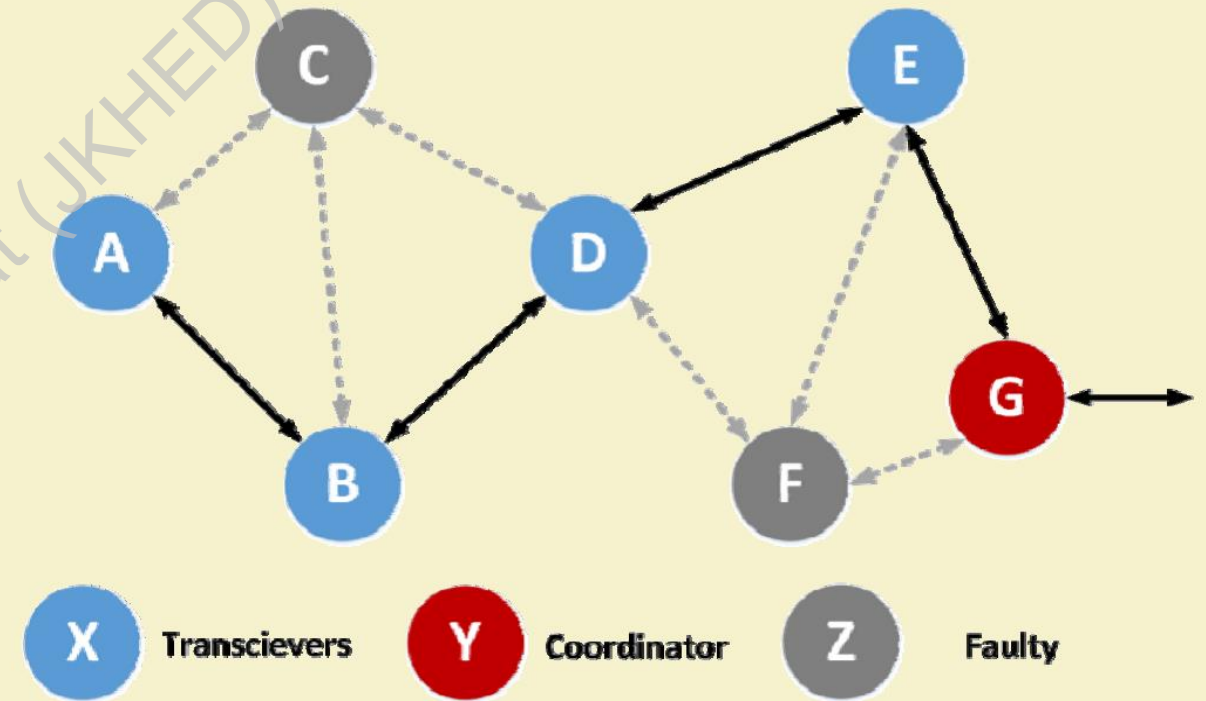
ZigBee Mesh

- ✓ In a mesh, any node can communicate with any other node within its range.
- ✓ If nodes are not in range, messages are relayed through intermediate nodes.
- ✓ This allows the network deployment over large areas.



ZigBee Mesh (Contd.)

- ✓ Meshes have increased network reliability.
- ✓ For example, if nodes C and F are down, the message packets from A can still be relayed to G via B and E.
- ✓ ZigBee mesh networks are self-configuring and self-healing.



Zigbee stack layers

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IEEE 802.15.4 and Zigbee

- The IEEE 802.15.4 standard defines the operation of low-rate wireless personal area networks (LR-WPANs).
- It specifies the physical layer (PHY) and media access control (MAC) for LR-WPANs, aiming to provide low-cost, low-speed ubiquitous communication between devices.
- Zigbee is closely associated with IEEE 802.15.4 as it builds upon this standard.
- While IEEE 802.15.4 defines the lower layers (PHY and MAC), Zigbee extends the standard by developing the upper layers which are not defined in IEEE 802.15.4, thus creating a full protocol stack for communication



Zigbee stack layers

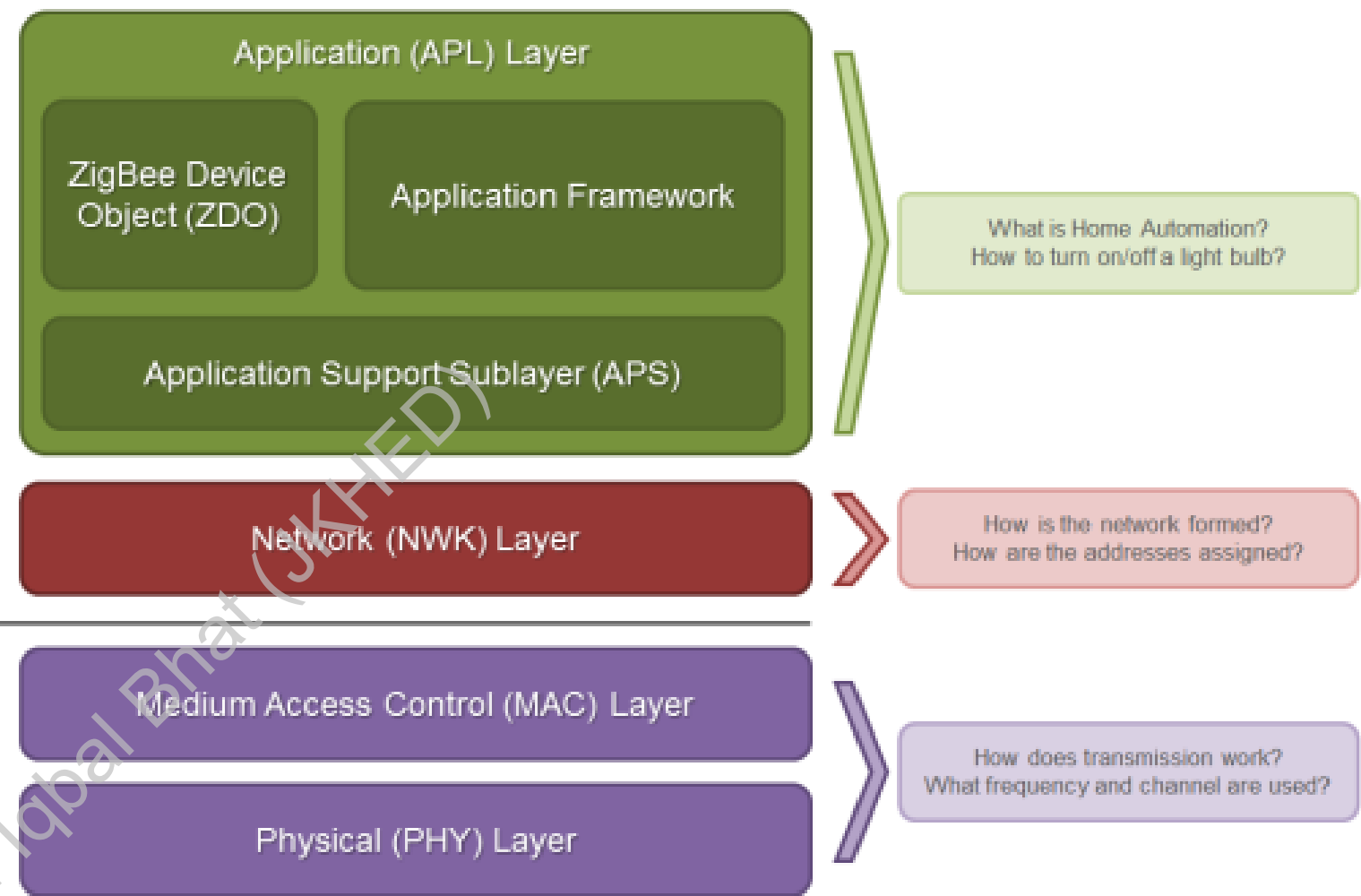
Important Components

ZDO

- ZigBee Device Object (Device management, Security, Policies)

APS

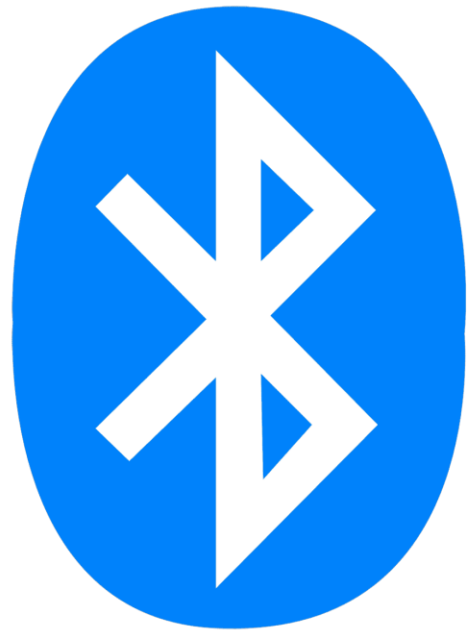
- Application Support Sub-layer (Interfacing and control services, bridge between network and other layers)



Zigbee layer	Descriptions
PHY	Defines the physical operation of the Zigbee device including receive sensitivity, channel rejection, output power, number of channels, chip modulation, and transmission rate specifications. Most Zigbee applications operate on the 2.4 GHz ISM band at a 250 kb/s data rate. See the IEEE 802.15.4 specification for details.
MAC	Manages RF data transactions between neighboring devices (point to point). The MAC includes services such as transmission retry and acknowledgment management, and collision avoidance techniques (CSMA-CA).
Network	Adds routing capabilities that allows RF data packets to traverse multiple devices (multiple hops) to route data from source to destination (peer to peer).
APS (AF)	Application layer that defines various addressing objects including profiles, clusters, and endpoints.
ZDO	Application layer that provides device and service discovery features and advanced network management capabilities.

Applications

- ✓ Building automation
- ✓ Remote control (RF4CE or RF for consumer electronics)
- ✓ Smart energy for home energy monitoring
- ✓ Health care for medical and fitness monitoring
- ✓ Home automation for control of smart homes
- ✓ Light Link for control of LED lighting
- ✓ Telecom services



Bluetooth®

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Introduction

- ✓ Bluetooth wireless technology is a short range communications technology.
- ✓ Intended for replacing cables connecting portable units
- ✓ Maintains high levels of security.
- ✓ Bluetooth technology is based on **Ad-hoc technology** also known as **Ad-hoc Piconets**.

Source: "[Wireless Communication - Bluetooth](#)", Tutorials Point (Online)

Features

- ✓ Bluetooth technology operates in the unlicensed industrial, scientific and medical (ISM) band at 2.4 to 2.485 GHz.
- ✓ Uses spread spectrum hopping, full-duplex signal at a nominal rate of 1600 hops/sec.
- ✓ Bluetooth supports 1Mbps data rate for version 1.2 and 3Mbps data rate for Version 2.0 combined with Error Data Rate.

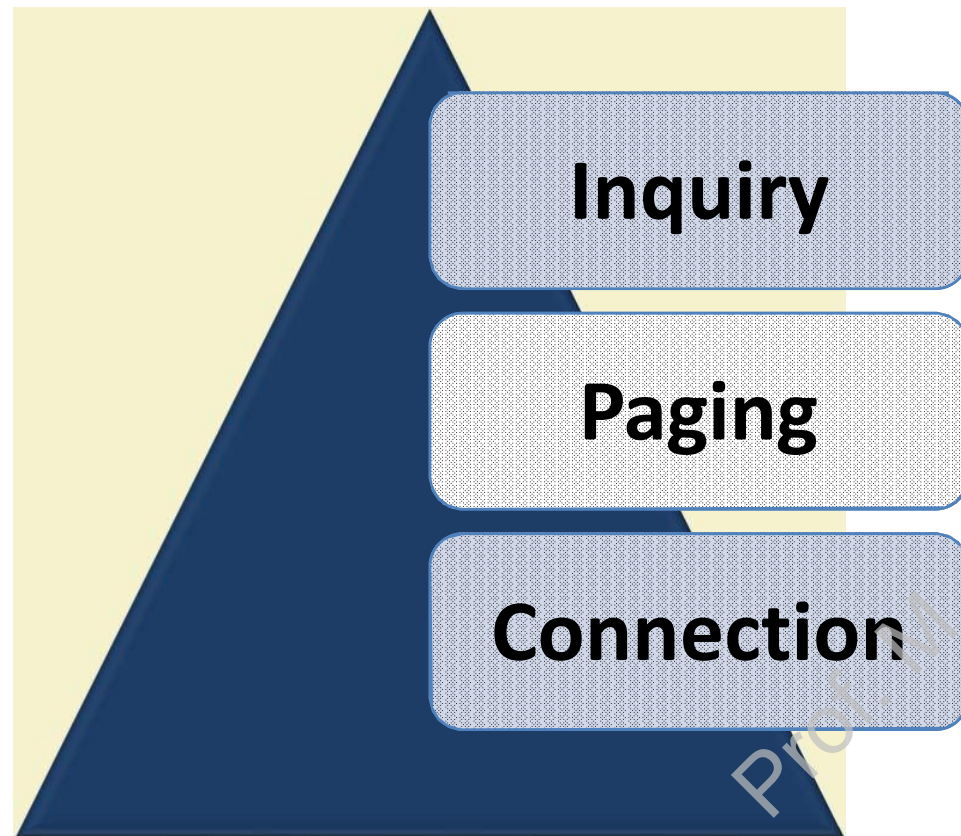
Source: "[Wireless Communication - Bluetooth](#)", Tutorials Point (Online)

Features

- ✓ Bluetooth operating range depends on the device:
 - **Class 3** radios have a range of up to 1 meter or 3 feet
 - **Class 2** radios are most commonly found in mobile devices have a range of 10 meters or 30 feet
 - **Class 1** radios are used primarily in industrial use cases have a range of 100 meters or 300 feet

Source: "[Wireless Communication - Bluetooth](#)", Tutorials Point (Online)

Connection Establishment



Inquiry run by one Bluetooth device to try to **discover other devices near it.**

Process of **forming a connection** between two Bluetooth devices.

A device either actively **participates** in the network or enters a low-power sleep mode.

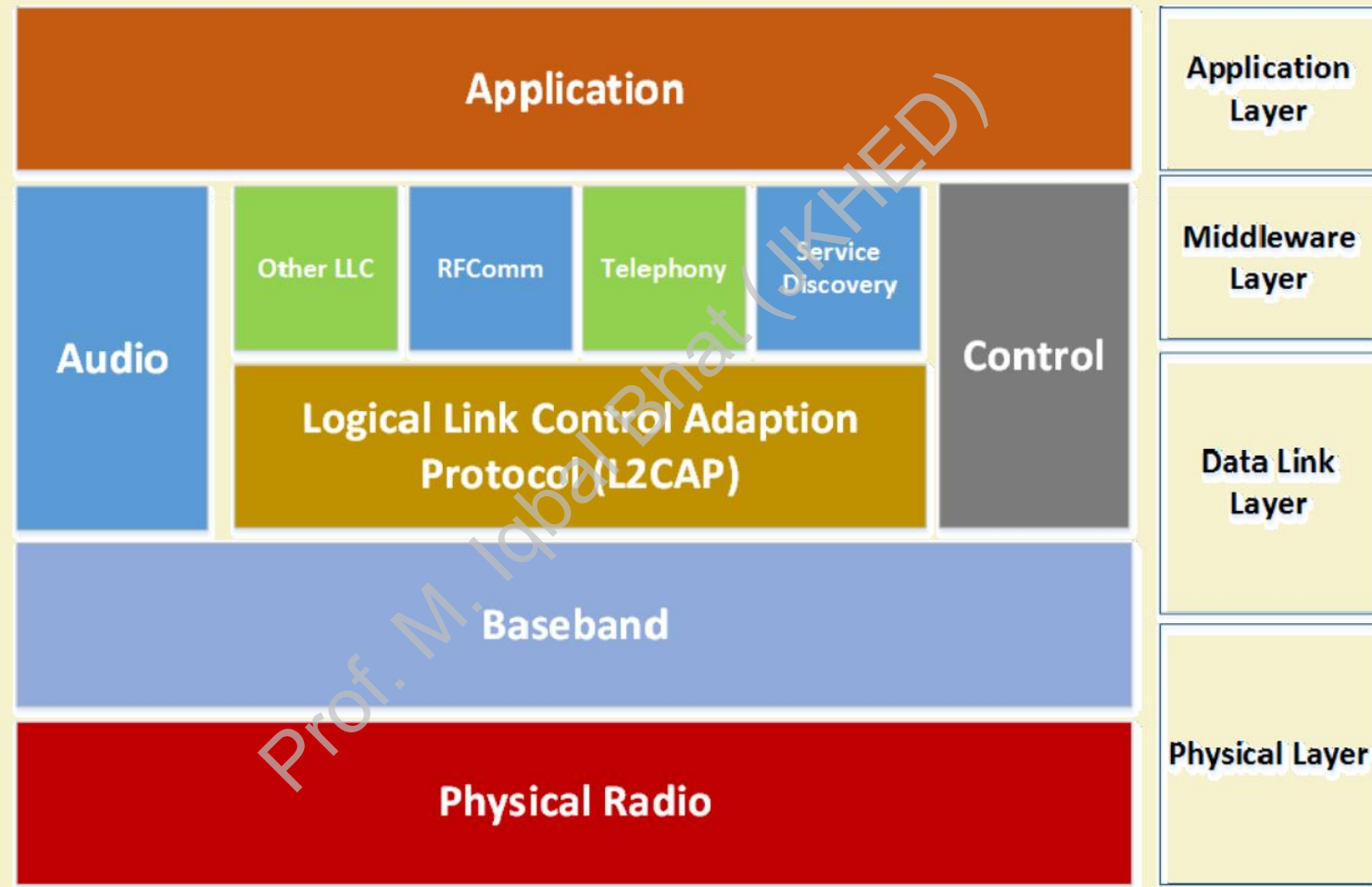
Source: "[Bluetooth Basics](#)", Tutorials, Sparkfun.com (Online)

Modes



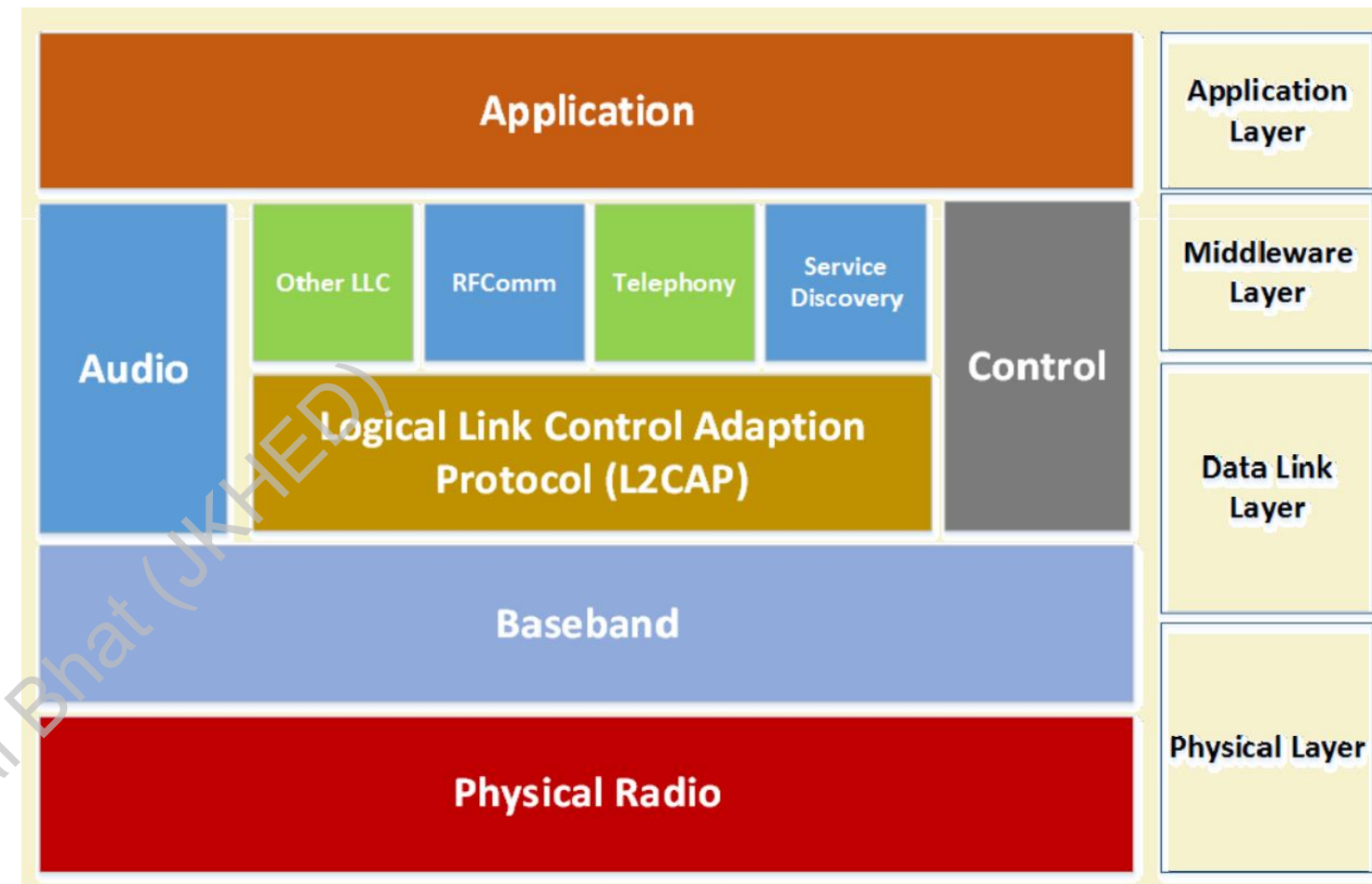
Source: "[Bluetooth Basics](#)", Tutorials, Sparkfun.com (Online)

Protocol Stack



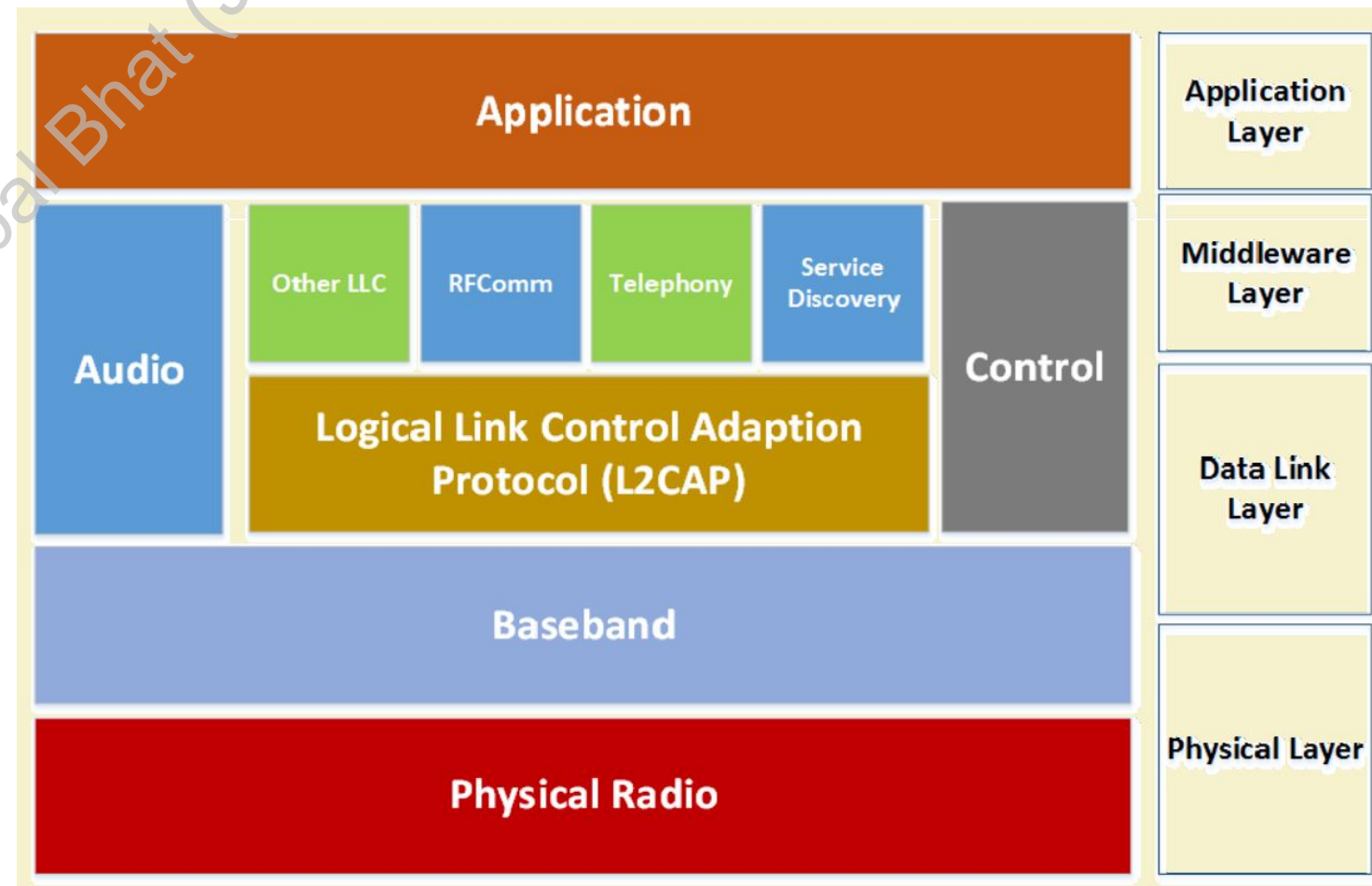
Baseband

- ✓ Physical layer of the Bluetooth.
- ✓ Manages physical channels and links.
- ✓ Other services include:
 - Error correction
 - Data whitening
 - Hop selection
 - Bluetooth security
- ✓ Manages asynchronous and synchronous links.
- ✓ Handles packets, paging and inquiry.



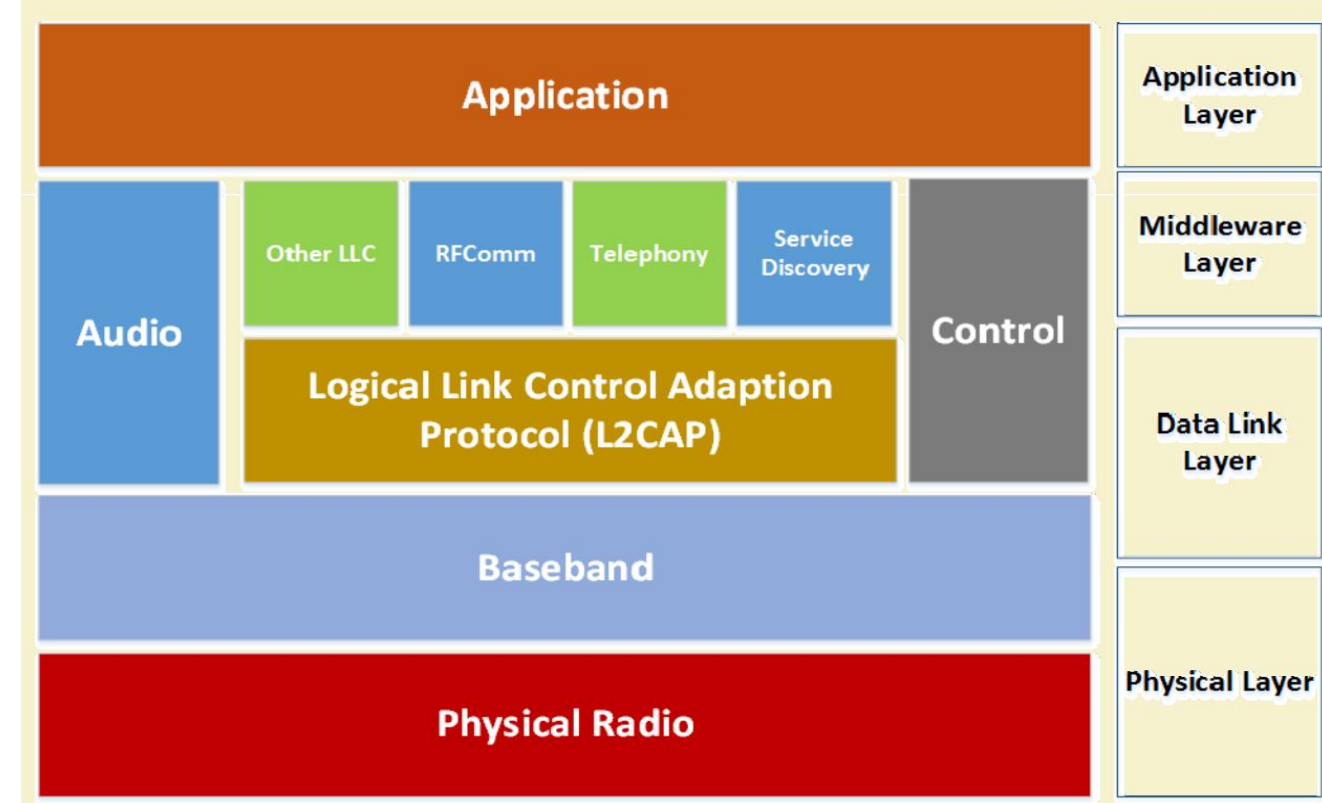
L2CAP

- ✓ The Logical Link Control and Adaptation Protocol (L2CAP).
- ✓ Layered over the Baseband Protocol and resides in the data link layer.
- ✓ Used to multiplex multiple logical connections between two devices.
- ✓ Provides connection-oriented and connectionless data services to upper layer protocols.
- ✓ Provides:
 - Protocol multiplexing capability
 - Segmentation and reassembly operation
 - Group abstractions



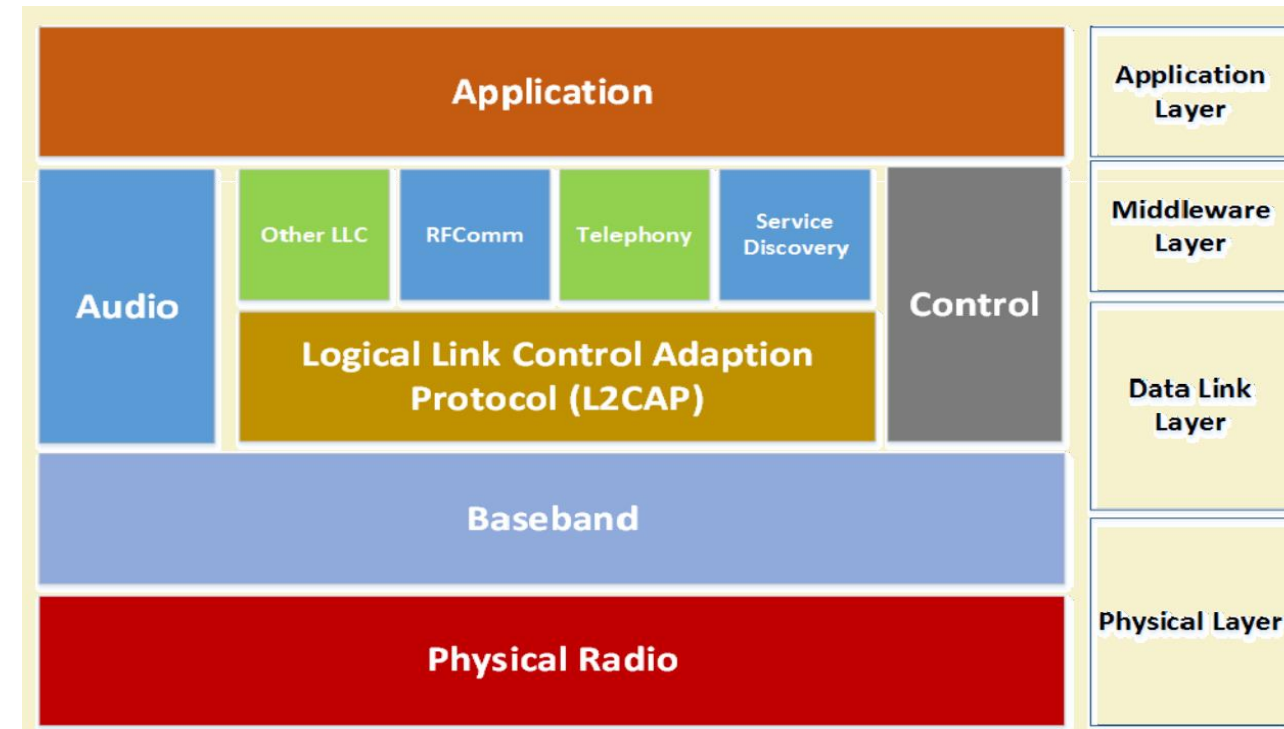
RFComm

- ✓ Radio Frequency Communications (RFCOMM).
- ✓ It is a cable replacement protocol used for generating a virtual serial data stream.
- ✓ RFCOMM provides for binary data transport .
- ✓ Emulates EIA-232 (formerly RS-232) control signals over the Bluetooth baseband layer, i.e. it is a serial port emulation.
- ✓ RFCOMM provides a simple reliable data stream to the user, similar to TCP.
- ✓ Supports up to 60 simultaneous connections between two BT devices.



Service Discovery Protocol (SDP)

- ✓ Enables applications to discover available services and their features.
- ✓ Addresses the unique characteristics of the Bluetooth environment such as, dynamic changes in the quality of services in RF proximity of devices in motion.
- ✓ Can function over a reliable packet transfer protocol.
- ✓ Uses a request/response model.



Piconets

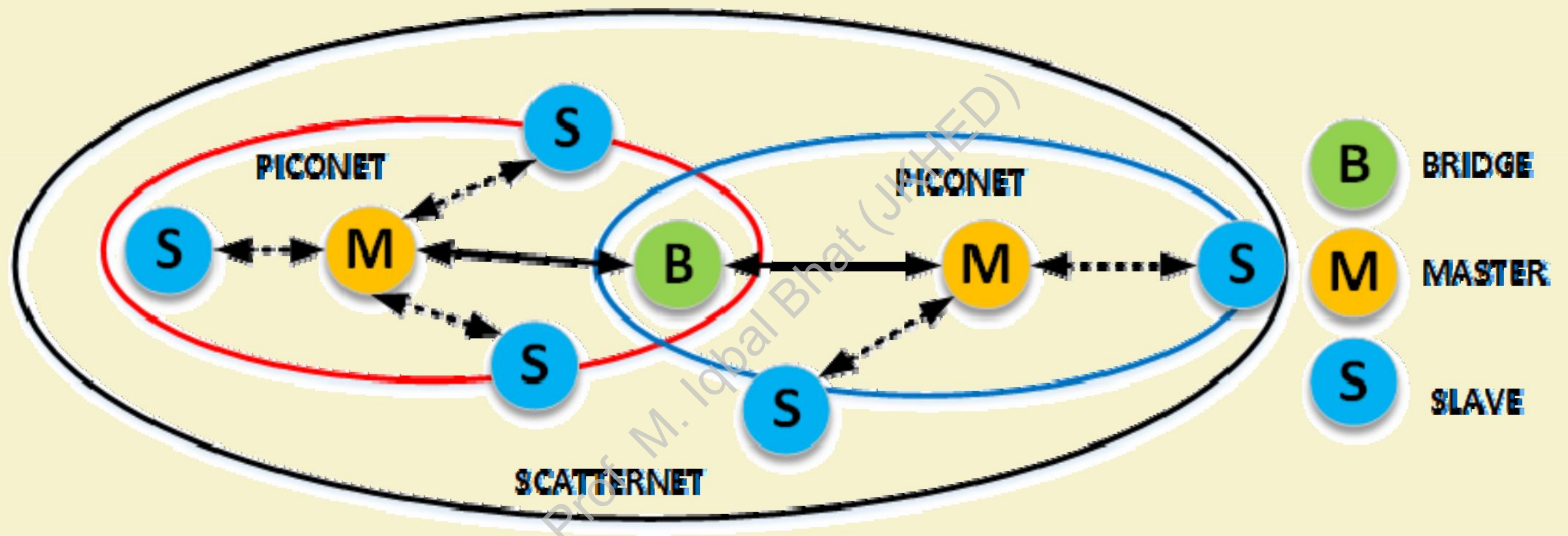
- ✓ Bluetooth enabled electronic devices connect and communicate wirelessly through short range networks known as **Piconets**.
- ✓ Bluetooth devices exist in small ad-hoc configurations with the ability to act either as master or slave.
- ✓ Provisions are in place, which allow for a **master** and a **slave** to switch their roles.
- ✓ That imples configuration is a point to point configuration with one master and one slave.

Source: "[Wireless Communication - Bluetooth](#)", Tutorials Point (Online)

Piconets

- ✓ When more than two Bluetooth devices communicate with one another, it is called a **PICONET**.
- ✓ A Piconet can contain up to seven slaves clustered around a single master.
- ✓ The device that initializes establishment of the Piconet becomes the **master**.
- ✓ The master is responsible for transmission control by dividing the network into a series of time slots amongst the network members, as a part of **time division multiplexing** scheme.

Source: "[Wireless Communication - Bluetooth](#)", Tutorials Point (Online)



Features of Piconet

- ✓ Within a Piconet, the clock and unique **48-bit address** of master determines the timing of various devices and the frequency hopping sequence of individual devices.
- ✓ Each Piconet device supports 7 simultaneous connections to other devices.
- ✓ Each device can communicate with several piconets simultaneously.
- ✓ Piconets are established dynamically and automatically as Bluetooth enabled devices enter and leave piconets.

Source: "[Wireless Communication - Bluetooth](#)", Tutorials Point (Online)

- ✓ There is no direct connection between the slaves.
- ✓ All connections are either master-to-slave or slave-to-master.
- ✓ Slaves are allowed to transmit once these have been polled by the master.
- ✓ Transmission starts in the slave-to-master time slot immediately following a polling packet from the master.
- ✓ A device can be a member of two or more Piconets.

Source: “[Wireless Communication - Bluetooth](#)”, Tutorials Point (Online)

- ✓ A device can be a slave in one Piconet and master in another. It however cannot be a master in more than once Piconets.
- ✓ Devices in adjacent Piconets provide a bridge to support inner-Piconet connections, allowing assemblies of linked Piconets to form a physically extensible communication infrastructure known as **Scatternet**.

Source: “[Wireless Communication - Bluetooth](#)”, Tutorials Point (Online)

Applications

- ✓ Audio players
- ✓ Home automation
- ✓ Smartphones
- ✓ Toys
- ✓ Hands free headphones
- ✓ Sensor networks

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BluetoothTM

Low Energy

Prof. M. Iqbal Bhat (JKHED)



BluetoothTM

Low Energy

Bluetooth Low Energy (BLE) is a wireless communication technology that enables low-power, short-range connectivity between devices. It offers a more efficient and cost-effective alternative to classical Bluetooth, making it ideal for a wide range of IoT and wearable applications.



Advantages of BLE over Classic Bluetooth

1

Lower Power Consumption

BLE devices can operate for months or years on a single coin cell battery, making them ideal for low-power IoT applications.

2

Faster Connection Establishment

BLE connections can be established in under 10 milliseconds, enabling real-time interactions and improved responsiveness.

3

Enhanced Efficiency

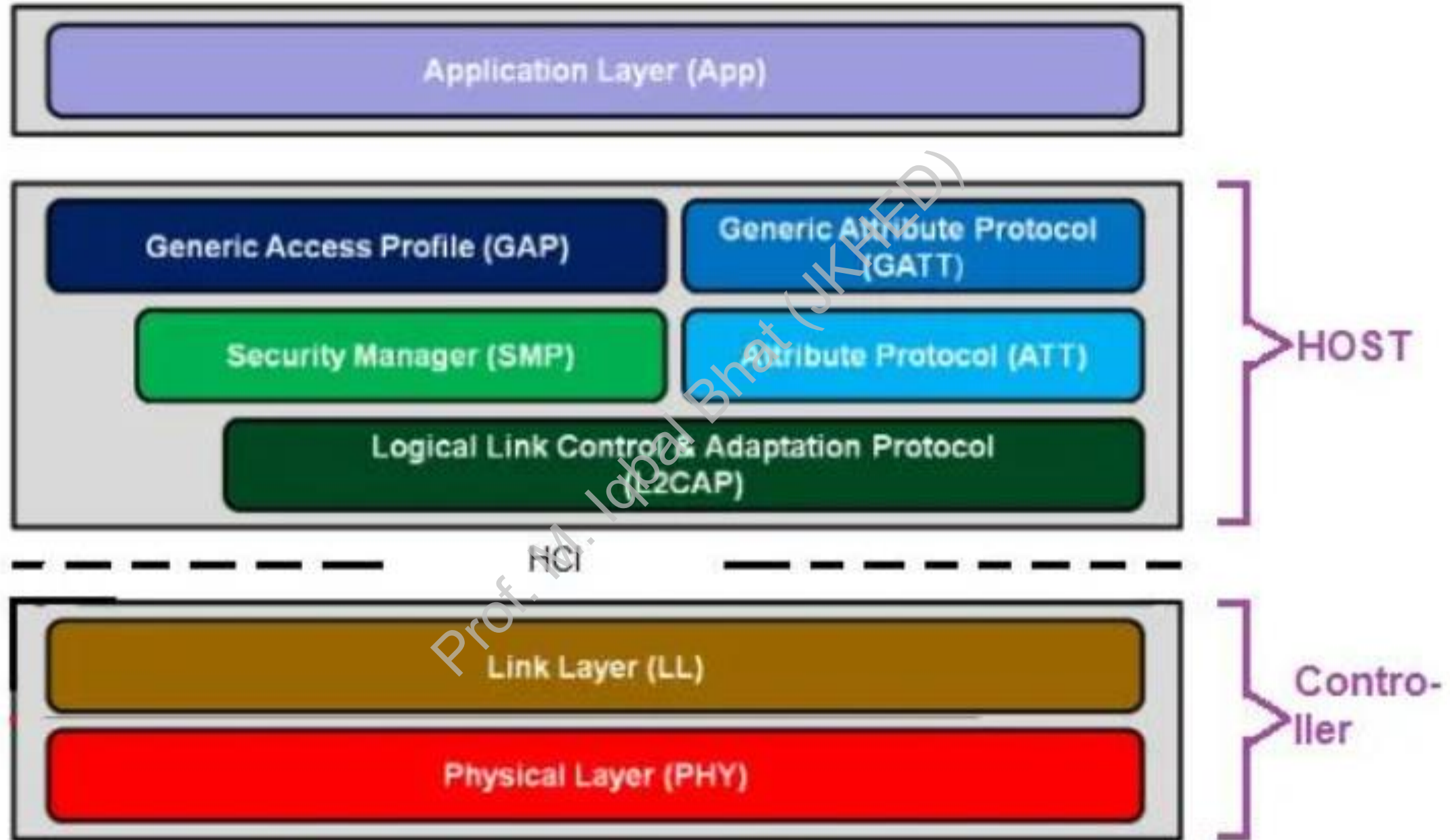
BLE uses a connectionless data model and adaptive frequency-hopping to reduce interference and improve throughput.

4

Broader Ecosystem

BLE is supported by a vast array of devices, from smartphones and wearables to smart home and industrial automation systems.

BLE Architecture and Components

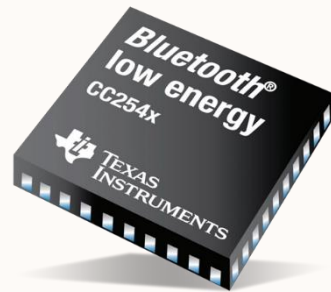


BLE (Bluetooth Low Energy) Protocol Stack

BLE Architecture and Components

BLE Chip

The core of a BLE device is the BLE chip, which integrates the radio, baseband, and processor on a single silicon die. This enables low-power operation and efficient wireless communication.



BLE Stack

The BLE stack consists of various layers, including the Physical Layer, Link Layer, Attribute Protocol, and Generic Attribute Profile. These work together to enable BLE's key features like low power, security, and interoperability.

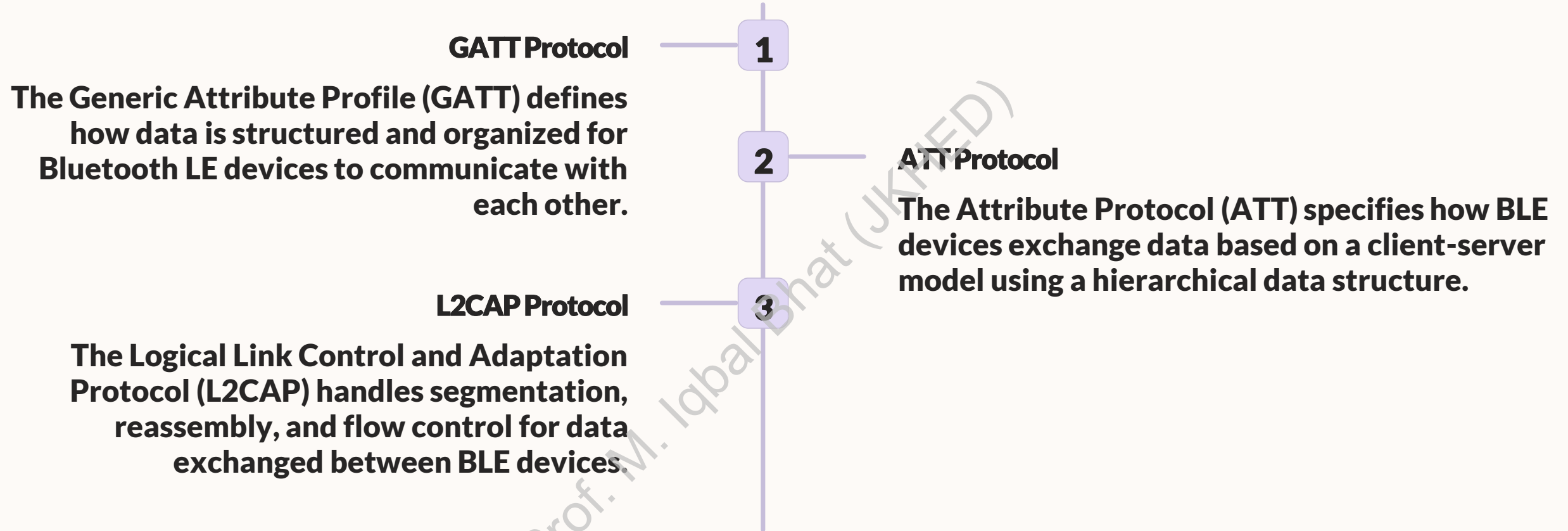
BLE Host

The BLE host is the software layer that runs on the main application processor, handling higher-level functionality like device discovery, connection management, and the application profile.

BLE Controller

The BLE controller is the hardware layer responsible for the low-level radio and link layer operations. It offloads time-critical tasks from the host to conserve power.

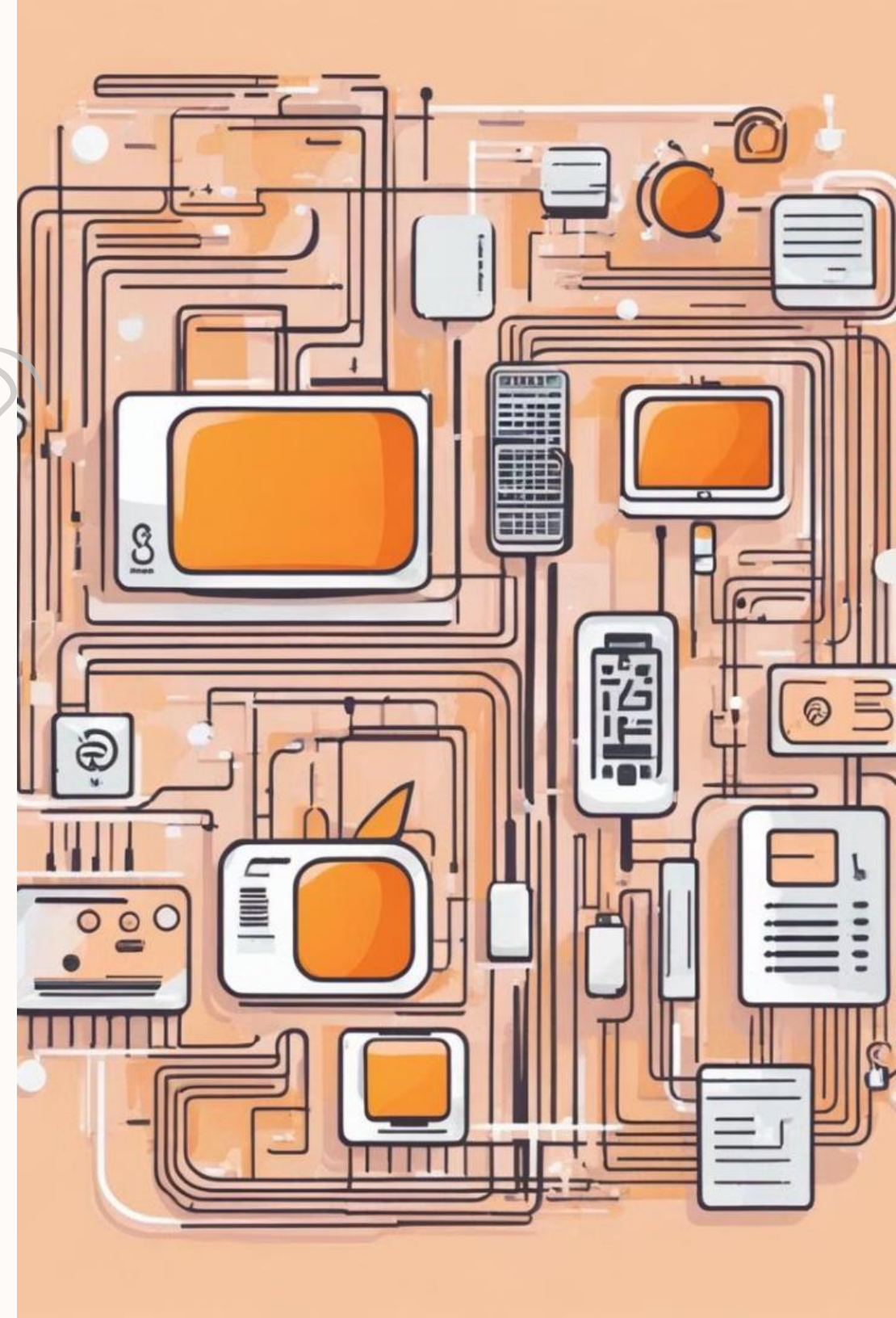
BLE Communication Protocols



BLE Advertising and Scanning

BLE devices use advertising to broadcast their presence and capabilities to nearby devices. Scanning allows other BLE devices to detect and connect to the advertising devices. This forms the foundation of BLE connectivity and discovery.

Advertisers broadcast small data packets containing information about the device, while scanners passively listen for listen for these broadcasts, allowing them to find and connect to connect to available BLE devices.





BLE Connections and Bonding

1

Connection Establishment

BLE devices initiate a connection by sending a connection request. Once accepted, a secure link is established between the devices.

2

Bonding and Pairing

Bonding involves exchanging encryption keys to create a trusted relationship between devices for future reconnections.

3

Connection Management

BLE devices can enter low-power modes and quickly reconnect when needed, optimizing battery life.

BLE Security and Encryption

Secure Pairing

BLE uses secure pairing methods like Passkey Entry and Out-of-Band (OOB) to establish trusted connections between devices, ensuring data is transmitted safely.

Encryption

BLE encrypts data using 128-bit AES-CCM, providing strong protection against eavesdropping and tampering during wireless communications.

Key Management

BLE devices securely manage encryption keys, renewing them regularly to maintain confidentiality and integrity of the data exchange.

Privacy Features

BLE supports device privacy through random address changes, preventing unauthorized tracking of BLE-enabled devices.

BLE Power Management and Optimization

Bluetooth Low Energy (BLE) is designed to be highly power-efficient, making it ideal for battery-powered IoT devices. Key power management features include low-power modes, connection interval optimization, and adaptive data rates.

2-5X

Lower

Power Consumption

10Y

Longer

Battery Life

BLE devices can enter low-power modes like Advertising, Scanning, and Connection modes to reduce energy consumption when not actively transmitting data. Connection interval optimization allows devices to adjust the frequency of data exchange, further reducing power draw.

BLE Applications and Use Cases



Smart Home

BLE enables low-power wireless connectivity for a variety of smart home devices, including lights, thermostats, security systems, and appliances, allowing for seamless integration and remote control.



Fitness Tracking

Wearable devices like fitness trackers and smartwatches leverage BLE to monitor user activity, activity, heart rate, sleep patterns, and other health metrics, providing valuable insights to promote an active lifestyle.



Medical Devices

BLE is used in various various medical devices, such as glucose glucose monitors, patient monitoring systems, and wireless wireless diagnostic tools, enabling remote remote monitoring and and data sharing to improve healthcare outcomes.



Location-based Services

BLE beacons can be deployed in retail stores, museums, and other public spaces to enable location-based services, services, such as proximity-based notifications, indoor navigation, and personalized marketing marketing experiences. experiences.

Bluetooth Classic vs BLE

Feature	Bluetooth Classic	Bluetooth Low Energy (BLE)
Power consumption	High (approx 1W)	Low (approx 0.01W-0.5W)
Communication Range	10m to 30m	10m to 30m
Data Rate	1Mbps for BR 2-3Mbps for EDR	500kbps-1Mbps
Modulation Technique	GFSK for BR 8-DPSK or $\pi/4$ -DQPSK for EDR	GFSK
RF Bandwidth	2.4 GHz ISM band (2400-2483.5 MHz)	2.4 GHz ISM band (2400-2483.5 MHz)
Number of Channels	79 channels each of width 1Mhz	40 channels each of width 2MHz
Spreading	Frequency Hopping Spread Spectrum (FHSS)	Frequency Hopping Spread Spectrum (FHSS)
Data link layer protocol	Time Division Multiple Access (TDMA)	Time Division Multiple Access (TDMA)
Error detection	8 bit CRC or 16 bit CRC, and ACKs	24 bit CRC, ACKs
Maximum number of active slaves	7	Unlimited

Conclusion

BLE technology is a key enabler for the Internet of Things, offering a low-power, cost-effective, and secure solution for short-range wireless communication. Its versatility and wide adoption make it a crucial technology for connecting and managing the growing number of devices in the IoT ecosystem.